ZyXEL XGS-4728F V4.00(BBC.0)C0

Release Note/Manual Supplement

Date: Apr. 8, 2011

This document describes the features in the XGS-4728F product for its 4.00(BBC.0)C0 release.

Support Platforms:

ZyXEL XGS-4728F V4.00(BBC.0)C0 supports models: ZyXEL XGS-4728F

Version:

OS Version: V4.00(BBC.0) | 04/08/2011
BootBase Version: V1.00 | 10/22/2007

Default Bootbase Setting:

ZyNOS Version	V4.00(BBC.0) | 04/08/2011 17:15:26
Bootbase Version	V1.00 | 10/22/2007 12:48:50
Serial Number	xxxxxxxxxxxxxx
Vendor Name	ZyXEL Communications Corp.
Product Model	XGS-4728F
ZyNOS Code Model	XGS-4728F
HTP Code Model	Unknown
ZyNOS ROM address	50080000
System Type	10
First MAC Address	0019CB000001
Last MAC Address	0019CB00001E
MAC Address Quantity	30
Default Country Code	FF
Boot Module Debug Flag	01
RomFile Version	23
RomFile Checksum	a64a
ZyNOS Checksum	797b
SNMP MIB level & OID	060102030405060708091011121314151617181920
Main Feature Bits	C0
Other Feature Bits	
01 20 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00-00 41 13 00 00 00	

Enhanced Features:

1. DHCPv6: client and relay
2. ICMPv6
3. IPv6 Path MTU
4. NDP: host and router
5. IPv6 address stateless auto-configuration: host and router
6. IPv6 static route
7. Guest VLAN
8. Password encryption

9. User access right
10. PPPoE IA and option 82
11. ECMP
12. Policy route
13. Configurable ARP learning mode
14. Recovery mechanism for error-disabled port/reason.
15. CPU protection
16. Support display IGMP snooping client
17. Support IPv6 in classifier
18. Remote configuration merge by TFTP
19. Support RFC4293 Management Information Base (MIB) for IP
20. Support RFC4292 IP Forwarding Table MIB
21. Support RFC4022 Management Information Base for the Transmission Control Protocol
22. Support RFC4113 Management Information Base for the User Datagram Protocol
23. CFM: Support both hostname and IP address as chassis ID
24. Synchronize system log and syslog
25. Configurable RSTP/MRSTP oper edge
26. CLI for ARP flush on specific IP or port lists
27. CLI for dump memory usage
28. CLI for RMON configuration
29. CLI/MIB for IGMP snooping report-proxy mode
30. Extend to 12 trunk groups
31. Extend to 128 routing domains
32. sFlow
33. VLAN counter
34. MLD snooping proxy
35. Extend TFTP file name length as 255 in MIB
36. Enhance IP source guard entries as 1024
37. MSTP operEdge option
38. Guest VLAN multi-secure number extends to 24

Bug Fix:

1. Port and MAC movement of IPv4 address.
2. Correct the calculation of Error Length packets.
3. Configure DHCP server via WEB will display error.
4. Enlarge the filename length of TFTP for SNMP.
5. The switch will not add outer SPVID tag on the incoming tagged DHCP frames when enable VLAN stacking and DHCP snooping.
6. Correct the calculation of Error packets on 10/12G port.
7. DUT will crash after creating a policy rule with a list of long name classifiers.
8. Remove one DHCP relay VLAN, all the others stop functioning

Known Issue:

1. Port movement of IPv6 address is not supported currently.
2. A filtering rule with discard source action will always filter packets regardless of source MAC or destination MAC.
3. When using three different spanning tree protocols (RSTP+MSTP+MRSTP) on the whole system, the system will loop.
4. When RSTP is enabled with ring topology, LACP can't be activated at the same ports.

Limitation of Settings:

1.	VLAN 1Q static entry	1024
2.	Static MAC forwarding entry	256
3.	MAC filtering entry	256
4.	Cluster member	24

5.	IP routing domain	128
6.	IGMP Filtering entry	256
7.	IGMP MVR entry	256
8.	VRRP entry	64
9.	OSPF area	4
10.	Protocol based VLAN entries per port	7
11.	DHCP Server	16
12.	Syslog server entry	4
13.	IP source guard entry	1024
14.	IP subnet based VLAN entry	16
15.	MVR VLAN entry	5
16.	VLAN-stacking Selective QinQ entry	768
17.	VLAN-mapping entry	768
18.	Private-VLAN entry	8

Change History:

V4.0(BBC.0)C0 (04/08/2011)
Support enhanced features & bug fix

V3.90(BBC.3)C0 (04/22/2010)
Revise FW version

V3.90(BBC.2)C0 (04/22/2010)
Support enhanced features & bug fix

V3.90(BBC.1)C0 (11/23/2009)
Support enhanced features

V3.90(BBC.0)C0 (10/15/2009)
Support enhanced features & bugs fix

V3.80(BBC.0)C0 (11/28/2007)
First Public Version

ZyXEL XGS-4728F V3.90(BBC.3)C0

Release Note/Manual Supplement

Date: Apirl 22, 2010

This document describes the features in the XGS-4728F product for its 3.90(BBC.3)C0 release.

Support Platforms:

ZyXEL XGS-4728F V3.90(BBC.3)C0 supports models: ZyXEL XGS-4728F

Version:

OS Version: V3.90(BBC.3) | 04/22/2010
BootBase Version: V1.00 | 10/22/2007

Default Bootbase Setting:

ZyNOS Version	V3.90(BBC.3) | 04/22/2010 09:40:31
Bootbase Version	V1.00 | 10/22/2007 12:48:50
Serial Number	xxxxxxxxxxxxxx
Vendor Name	ZyXEL Communications Corp.
Product Model	XGS-4728F
ZyNOS Code Model	XGS-4728F
HTP Code Model	Unknown
ZyNOS ROM address	50080000
System Type	10
First MAC Address	0019CB000001
Last MAC Address	0019CB00001E
MAC Address Quantity	30
Default Country Code	FF
Boot Module Debug Flag	01
RomFile Version	DB
RomFile Checksum	9d5e
ZyNOS Checksum	9502
SNMP MIB level & OID	060102030405060708091011121314151617181920
Main Feature Bits	C0
Other Feature Bits	
01 20 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00-00 41 13 00 00 00	

Enhanced Features:

None

Bug Fix:

1. Revise FW version

Known Issue:

1.Port movement of IP address is not supported currently.
2.A filtering rule with discard source action will always filter packets regardless of source MAC or destination MAC.
3.It may cause CPU overload when create too many MSTP instance or join too many ports of each MSTP instance
4. When using three different spanning tree protocols (RSTP+MSTP+MRSTP) on the whole system, the system will loop.

Limitation of Settings:

1.	VLAN 1Q static entry	1024
2.	Static MAC forwarding entry	256
3.	MAC filtering entry	256
4.	Cluster member	24
5.	IP routing domain	64
6.	IGMP Filtering entry	256
7.	IGMP MVR entry	256
8.	VRRP entry	64
9.	OSPF area	4
10.	Protocol based VLAN entries per port	7
11.	DHCP Server	16
12.	Syslog server entry	4
13.	IP souce guard entry	128
14.	IP subnet based VLAN entry	16
15.	MVR VLAN entry	5
16.	Vlan-stacking Selective QinQ entry	768
17.	Vlan-mapping entry	768
18.	Private-vlan entry	8

Change History:

V3.90(BBC.3)C0 (04/22/2010)
Revise FW version

V3.90(BBC.2)C0 (04/22/2010)
Support enhanced features & bug fix

V3.90(BBC.1)C0 (11/23/2009)
Support enhanced features

V3.90(BBC.0)C0 (10/15/2009)
Support enhanced features & bugs fix

V3.80(BBC.0)C0 (11/28/2007)
First Public Version

ZyXEL XGS-4728F V3.90(BBC.2)C0

Release Note/Manual Supplement

Date: Apirl 22, 2010

This document describes the features in the XGS-4728F product for its 3.90(BBC.2)C0 release.

Support Platforms:

ZyXEL XGS-4728F V3.90(BBC.2)C0 supports models: ZyXEL XGS-4728F

Version:

OS Version: V3.90(BBC.2) | 04/22/2010
BootBase Version: V1.00 | 10/22/2007

Default Bootbase Setting:

ZyNOS Version	V3.90(BBC.2) | 04/22/2010 09:33:35
Bootbase Version	V1.00 | 10/22/2007 12:48:50
Serial Number	xxxxxxxxxxxxxx
Vendor Name	ZyXEL Communications Corp.
Product Model	XGS-4728F
ZyNOS Code Model	XGS-4728F
HTP Code Model	Unknown
ZyNOS ROM address	50080000
System Type	10
First MAC Address	0019CB000001
Last MAC Address	0019CB00001E
MAC Address Quantity	30
Default Country Code	FF
Boot Module Debug Flag	01
RomFile Version	DB
RomFile Checksum	9d5e
ZyNOS Checksum	510b
SNMP MIB level & OID	060102030405060708091011121314151617181920
Main Feature Bits	C0
Other Feature Bits	
01 20 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00-00 41 13 00 00 00	

Enhanced Features:

1. Smart Isolation
2. Private vlan
3. Classifier can qualify destination-port
4. Hostname prompt cli truncated to 10 chars, should be minimum 20
5. Release the upper limit of mac aging time (10-1000000)
6. show mac trunk id

Bug Fix:

1. [RSTP]The DUT will looping when send multicast data to XGS ring topology
2. [RSTP]The blocking port cannot be fixed when no port link down or changing configure over night.
3. [RSTP] The port link down but port state of spanning tree still forwarding after overnight topology change.
4. Trunk flush arp sometimes will cause ARP table does not sync with hardware table
5. Correct BPDU control range to follow 802.1d
6. vlan-stacking and port-security IOP issue
7. The system time will be delayed 15minutes every 24 hours after sync up with NTP server
8. igmp snooping query loss issue
9. The IGMP multicast service will display strange up time value after using show command. The CLI command on ZyNOS F/W is “show multicast”
10. When issuing the command "wr memory" packet loss is seen when pinging the out and inband management address
11. IPSG - filter illegal smac out
12. OSPF does not delete invalid route from routing table.
13. The MSTP can not work when our devices cooperate with HUAWEI device. This is an interoperability issue
14. dhcp discover packet sometimes will pass through when enable port-security and port-isolation at the same time
15. The DUT login session will hang up by Second tacacs+ or radius server when tacacs+ server timeout = 1
16. The DHCP relay-broadcast cannot work
17. DUT will crash when sending LLDP trap and show logging
18. Correct BPDU control range.
19. mac-aging time does not work under certain configuration

Known Issue:

1.Port movement of IP address is not supported currently.
2.A filtering rule with discard source action will always filter packets regardless of source MAC or destination MAC.
3.It may cause CPU overload when create too many MSTP instance or join too many ports of each MSTP instance
4. When using three different spanning tree protocols (RSTP+MSTP+MRSTP) on the whole system, the system will loop.

Limitation of Settings:

1.	VLAN 1Q static entry	1024
2.	Static MAC forwarding entry	256
3.	MAC filtering entry	256
4.	Cluster member	24
5.	IP routing domain	64
6.	IGMP Filtering entry	256
7.	IGMP MVR entry	256
8.	VRRP entry	64
9.	OSPF area	4
10.	Protocol based VLAN entries per port	7
11.	DHCP Server	16
12.	Syslog server entry	4
13.	IP souce guard entry	128
14.	IP subnet based VLAN entry	16
15.	MVR VLAN entry	5

16.	Vlan-stacking Selective QinQ entry	768
17.	Vlan-mapping entry	768
18.	Private-vlan entry	8

Change History:

V3.90(BBC.2)C0 (04/22/2010)
Support enhanced features & bug fix

V3.90(BBC.1)C0 (11/23/2009)
Support enhanced features

V3.90(BBC.0)C0 (10/15/2009)
Support enhanced features & bugs fix

V3.80(BBC.0)C0 (11/28/2007)
First Public Version

ZyXEL XGS-4728F V3.90(BBC.1)C0

Release Note/Manual Supplement

Date: November 23, 2009

This document describes the features in the XGS-4728F product for its 3.90(BBC.1)C0 release.

Support Platforms:

ZyXEL XGS-4728F V3.90(BBC.1)C0 supports models: ZyXEL XGS-4728F

Version:

OS Version: V3.90(BBC.1) | 11/23/2009
BootBase Version: V1.01 | 05/22/2007

Default Bootbase Setting:

ZyNOS Version	V3.90(BBC.1) | 11/23/2009 16:26:00
Bootbase Version	V1.00 | 10/22/2007 12:48:50
Serial Number	xxxxxxxxxxxxxx
Vendor Name	ZyXEL Communications Corp.
Product Model	XGS-4728F
ZyNOS Code Model	XGS-4728F
HTP Code Model	Unknown
ZyNOS ROM address	50080000
System Type	10
First MAC Address	0019CB000001
Last MAC Address	0019CB00001E
MAC Address Quantity	30
Default Country Code	FF
Boot Module Debug Flag	01
RomFile Version	DB
RomFile Checksum	9d5e
ZyNOS Checksum	cf6f
SNMP MIB level & OID	060102030405060708091011121314151617181920
Main Feature Bits	C0
Other Feature Bits	
01 20 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00-00 41 13 00 00 00	

Enhanced Features:

1. Support new MAC prefix

Bug Fix:

None

Known Issue:

1. Port movement of IP address is not supported currently.
2. A filtering rule with discard source action will always filter packets regardless of source MAC or destination MAC.
3. It may cause CPU overload when create too many MSTP instance or join too many ports of each MSTP instance
4. When using three different spanning tree protocols (RSTP+MSTP+MRSTP) on the whole system, the system will loop.

Limitation of Settings:

1.	VLAN 1Q static entry	1024
2.	Static MAC forwarding entry	256
3.	MAC filtering entry	256
4.	Cluster member	24
5.	IP routing domain	64
6.	IGMP Filtering entry	256
7.	IGMP MVR entry	256
8.	VRRP entry	64
9.	OSPF area	4
10.	Protocol based VLAN entries per port	7
11.	DHCP Server	16
12.	Syslog server entry	4
13.	IP souce guard entry	128
14.	IP subnet based VLAN entry	16
15.	MVR VLAN entry	5
16.	Vlan-stacking Selective QinQ entry	768
17.	Vlan-mapping entry	768

Change History:

V3.90(BBC.1)C0 (11/23/2009)
Support enhanced features

V3.90(BBC.0)C0 (10/15/2009)
Support enhanced features & bugs fix

V3.80(BBC.0)C0 (11/28/2007)
First Public Version

ZyXEL XGS-4728F V3.90(BBC.0)C0

Release Note/Manual Supplement

Date: October 15, 2009

This document describes the features in the XGS-4728F product for its 3.90(BBC.0)C0 release.

Support Platforms:

ZyXEL XGS-4728F V3.90(BBC.0)C0 supports models: ZyXEL XGS-4728F

Version:

OS Version: V3.90(BBC.0) | 10/15/2009
BootBase Version: V1.00 | 05/22/2007

Default Bootbase Setting:

ZyNOS Version	V3.90(BBC.0) | 10/15/2009 18:04:22
Bootbase Version	V1.00 | 10/22/2007 12:48:50
Serial Number	xxxxxxxxxxxxxx
Vendor Name	ZyXEL Communications Corp.
Product Model	XGS-4728F
ZyNOS Code Model	XGS-4728F
HTP Code Model	Unknown
ZyNOS ROM address	50080000
System Type	10
First MAC Address	0019CB000001
Last MAC Address	0019CB00001E
MAC Address Quantity	30
Default Country Code	FF
Boot Module Debug Flag	01
RomFile Version	DB
RomFile Checksum	9d5e
ZyNOS Checksum	5868
SNMP MIB level & OID	060102030405060708091011121314151617181920
Main Feature Bits	C0
Other Feature Bits	
01 20 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00-00 41 13 00 00 00	

Enhanced Features:

1. LLDP
2. VLAN Mapping
3. Selective QinQ
4. VLAN MAC Limit
5. TACACS+ authorization
6. IGMP Throttling
7. L2PT

8. MAC-freeze GUI interface
9. LACP algorithm of Source/Destination
10. MAC searching feature
11. VLAN search page
12. AAA configuration convert
13. Loopback support size and count parameters
14. Support IGMP snooping fast leave
15. Support IGMP snooping statistics and MIB
16. Support DDM MIB(private MIB)
17. Support 802.3ah MIB
18. Extend MVR to 5 multicast VLANs
19. Revised CFM private MIB, and include MIB file for LBM and LTM
20. Change web privilege
21. OSPF summary
22. Extend VRRP to 64 entries
23. VLAN counters
24. Support memory statistics

Known Issue:

1. Port movement of IP address is not supported currently.
2. A filtering rule with discard source action will always filter packets regardless of source MAC or destination MAC.
3. It may cause CPU overload when create too many MSTP instance or join too many ports of each MSTP instance
4. When using three different spanning tree protocols (RSTP+MSTP+MRSTP) on the whole system, the system will loop.

Bug Fix:

9. VLAN stacking will be malfunctioned if the last port configured as “Normal” port.
10. In the case of unreachable gateway, switch would become unreachable for a while.
11. Configuration conflict will be happened on loopguard and MRTSP.
12. The action of changing default gateway will not take effect unless reboot.
13. For a few special secure client applications, the switch would become unreachable if user key-in different usernames for login in the same session.
14. In some cases of multicast group joins/leaves, it may cause switch crash.

Limitation of Settings:

1.	VLAN 1Q static entry	1024
2.	Static MAC forwarding entry	256
3.	MAC filtering entry	256
4.	Cluster member	24
5.	IP routing domain	64
6.	IGMP Filtering entry	256
7.	IGMP MVR entry	256
8.	VRRP entry	64
9.	OSPF area	4
10.	Protocol based VLAN entries per port	7
11.	DHCP Server	16
12.	Syslog server entry	4
13.	IP souce guard entry	128
14.	IP subnet based VLAN entry	16
15.	MVR VLAN entry	5
16.	Vlan-stacking Selective QinQ entry	768
17.	Vlan-mapping entry	768

Change History:

V3.90(BBC.0)C0 (10/15/2009)
Support enhanced features & bugs fix

V3.80(BBC.0)C0 (11/28/2007)
First Public Version

ZyXEL XGS-4728F V3.80(BBC.0)C0

Release Note/Manual Supplement

Date: Nov 28, 2007

This document describes the features in the XGS-4728F product for its 3.80(BBC.0)C0 release.

Support Platforms:

ZyXEL XGS-4728F V3.80(BBC.0)C0 supports models: ZyXEL XGS-4728F

Version:

OS Version: V3.80(BBC.0)C0 | 11/28/2007
BootBase Version: V1.00 | 10/22/2007

Default Bootbase Setting:

OS Version	V3.80(BBC.0) | 11/28/2007 23:40:47
Bootbase Version	V1.00 | 10/22/2007 12:48:50
Vendor Name	ZyXEL
Product Model	XGS-4728F
OS Code Model	XGS-4728F
HTP Code Model	Unknown
OS ROM address	50080000
System Type	10
MAC Address	0019CB000001
Default Country Code	FF
Boot Module Debug Flag	01
RomFile Version	D5
RomFile Checksum	67b5
OS Checksum	a7c2
SNMP MIB level & OID	060102030405060708091011121314151617181920
Main Feature Bits	C0
Other Feature Bits	
01 20 00 00 00 00 00 00-00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00-00 41 13 00 00 00	

Features:

1. 24 ports Auto MDI/MDI-X 1000Mbps compliant with IEEE802.3u/x/ad
2. 24 ports Mini GBIC 1000Mbps compliant with IEEE802.3x/ab
3. 1 uplink module with 2 XFP ports
4. 2 10G ports for stacking
5. MAC address learning
6. Support IEEE 802.1D transparent bridge
7. Support IEEE 802.1Q tagged VLAN
8. Support Port-based VLAN
9. Support IEEE 802.1X

10. Support IEEE 802.1W
11. Support IEEE 802.3AD
12. Support GVRP
13. IGMP snooping
14. Support IEEE 802.1p
15. Automatic age out
16. Specific MAC addresses forwarding per port
17. MAC Filtering
18. Port Trunking
19. Port Mirroring
20. Filtering/Mirroring by L2/L3/L4 rules
21. Bandwidth control by L2/L3/L4 rules
22. Egress traffic shaping per port
23. Broadcast Storm Control
24. Support Strict Priority and WRR queuing method
25. Static IP management or dynamic IP(DHCP client)
26. Multiple IP address management
27. Firmware upgrade and configuration backup/restore.
28. Remote manageable.
29. Cluster Management
30. WEB manageable
31. Support RFC-1213 MIB II
32. Support RFC-1493 Bridge MIB
33. Support RFC-1643 Ethernet MIB
34. Support RFC-1757 Four group of RMON
35. Support RFC-2674 VLAN MIB
36. NEW CLI supported
37. System configuration backup / restore in text-mode
38. Multiple login supported
39. Radius management authentication
40. Hardware Monitor
41. External power backup for internal power module failure
42. SSH v1/v2
43. SSL
44. Multicast
45. MVR
46. IGMP Filtering
47. CIR & PIR
48. MAC Freeze
49. Intrusion Lock
50. Support private MIB
51. Support Syslog
52. Protocol-based VLAN
53. MRSTP
54. Multicast routing on MVR
55. Administration user management
56. RADIUS enhancement (for 802.1x enhancement and multiple radius servers)
57. IGMP enhancement (timer and priority)
58. Web enhancement (Select all and Clone)
59. OSPF enhancement (passive interface and interface priority)
60. 802.1s MSTP
61. SNMPv3
62. enlarge syslog buffer
63. 802.3ah OAM
64. Multilevel CLI
65. SNMP trap group
66. configurable multicast VLAN
67. independent IGMP snooping and MVR setting
68. enlarge concurrent telnet session
69. IP source guard(Static binding, DHCP snooping, ARP inspection)

70. Two rate three color marking
71. IP subnet VLAN
72. RADIUS MAC login
73. RADIUS accounting
74. TACACS+ authentication
75. TACACS+ accounting
76. IGMPv3
77. IGMPv3 snooping
78. IGMP port-based
79. Loop guard
80. Daylight saving

Known Issue:

1. Port movement of IP address is not supported currently.
2. A filtering rule with discard source action will always filter packets regardless of source MAC or destination MAC.

Bug Fix:

Firmware Upgrade:

The XGS-4728F uses FTP to upgrade firmware in run-time through its built-in FTP server. You can use any FTP client (for example, ftp.exe in Windows) to upgrade XGS-4728F. The upgrade procedure is as follows:

Upgrade XGS-4728F FW:

```
C:\> ftp <XGS-4728F IP address>
User : <Enter>
Password: 1234
230 Logged in
ftp> put 380BBC0C0.bin ras
ftp> bye
```

Where

- User name : just press <Enter>
- Password : the management password, 1234 by default
- 380BBC0C0.bin : the name of firmware file you want to upgrade
- ras : the internal firmware name in XGS-4728F